

# Market Structures and the Distribution of Content<sup>\*</sup>

Matthew J. H. Murphy<sup>†</sup>

This version: May 5, 2025

I study how market structure influences the distribution of entertainment products. When producers compete for attention, ex-ante uncertainty over their quality causes producers to inefficiently sort across horizontally differentiated products (“genres”). In the competitive equilibrium, production is inefficiently concentrated in high-demand, homogeneous, genres. Relative to when producers can price-discriminate based on quality, flat pricing alleviates this inefficiency when genres are ex-ante similar in quality, but aggravates this inefficiency when genres are ex-ante different. Merging producers into an integrated platform restores efficiency. Recommendation systems are able to sometimes improve on the competitive equilibrium by selectively suppressing content.

---

<sup>\*</sup>Thanks.

<sup>†</sup>University of Pennsylvania. Email: [mjhm@sas.upenn.edu](mailto:mjhm@sas.upenn.edu)

# 1 Introduction

Digitization has led to a radical change in how consumers experience cultural products. Historically, consumers purchased entertainment products (or experiences) as they would any other consumption good: To watch a movie, a consumer would go to a movie theatre and buy a ticket to that specific show; To listen to an album, a consumer would go to a record store and purchase the album. The rise of streaming services has radically changed these market interactions. Instead of purchasing individual pieces of content, consumers are now able to purchase access to a bundle of content and consume any content in the bundle at zero marginal price. A natural question is how these changes in market structure have influence the types of products consumed, and hence the types of products created.

Existing in parallel to these changes in market structure is a confluence of unique product characteristics. As in many industries, there is a large degree of horizontal differentiation across cultural products, as well as pronounced vertical differentiation. Striking in specifically this industry is the asymmetry in these dimensions of differentiation. A producer intentionally decides what type of song to write, or what genre of movie to produce. However, a producer cannot perfectly determine the quality of the work they produce. Previous work has emphasized the significance of ex-ante uncertainty over quality in these industries. For instance, in the music industry, Aguiar and Waldfogel (2018) provide evidence that quality is largely unpredictable. In the movie industry, the mantra of “Nobody Knows Anything” about which movies will perform well, is prevalent.<sup>1</sup> If producers aim to maximize profits, they must incorporate stochastic quality into their decisions about what types of products to produce.

I study how these fundamental changes in market structure *interact* with uncertainty over quality and horizontal producer sorting. The model I construct highlights unique market failures due to quality unpredictability, and shows how changes in the underlying market structure influence these market failures. In particular, my model highlights how a decentralized competitive equilibrium leads to an inefficient lack of diversity in production. I argue that centralizing production through an integrated

---

<sup>1</sup>This phrase comes from famous screenwriter William Goldman. See Hennig-Thurau, Houston, et al. (2019) for a discussion of the history of the mantra and its impact on the industry.

platform (e.g., Netflix) can alleviate this inefficiency. I also show that, even if production remains decentralized, there is scope for a recommendation system (e.g., Spotify) to improve social welfare influence production decisions by suppressing some content.

In my model, producers, uncertain about their own quality, choose what “genre” of product to produce in order to maximize their expected profits. This is the horizontal dimension of differentiation. After producers make their production decisions, their quality is realized and perfectly observed by consumers.<sup>2</sup> On the other side of the market are consumers, heterogeneous over both which genre from which they consume content, and *how much* of these products they are willing to consume. This second dimension of consumer preferences acts as an upper bound on the quantity of consumption (consumers are heterogeneous in how much time they are able to spend watching TV or listening to music). Given this decomposition of consumer types, consumers will purchase those products which given them the highest net utility, until either no positive-utility products remain or they reach their capacity constraint.

My first main result shows that ex-ante uncertainty is both necessary and sufficient for producer sorting to be inefficient in a decentralized market. In particular, if producers knew their quality before they sorted into genres, the markets would operate efficiently. In this case, producers with the highest realized quality will sort into the genres with the highest demand, since they face no competition. It is also efficient for them to enter this genre, since it is socially beneficial for the most consumers to consume the best content. This process continues in descending order across the realization of quality, so that there is efficient sorting. However, if producers are uncertain about their quality, their sorting inflicts an externality on other firms. I show that, in this setting, merging producers into a single integrated platform alleviates the inefficiency due to producer sorting. This suggests that as the shift to streaming has led to an integration of producers and distributors, the distribution of content produced should become more efficient.

My second main result characterizes the nature of this inefficiency. In particular, when firms are uncertain about their quality, they concentrate too heavily in high-

---

<sup>2</sup>Hence, there is no ex-post uncertainty over quality, only ex-ante uncertainty. This distinguishes my model from previous studies on recommendation systems, which focus on platform’s ability to improve welfare through providing information and decreasing search costs.

demand genres. This means that the competitive equilibrium features too much concentration relative to the socially optimal level. I additionally show that the more that the prevailing market price depends on realized product quality, the larger this concentration. Thus, the flat pricing widely observed in these industries encourages product diversity.<sup>3</sup>

After characterizing this market failure, I show how distribution of consumer demand and quality uncertainty affect the distribution of content in the different market structures. Holding total demand constant, I show that genres with a larger proportion of “super-fans” always receive fewer producers in the competitive equilibrium. When the total mass of producers is small, this also holds in the centralized allocation of producers (the integrated platform’s allocation). In contrast, when producers are abundant, genres with more super-fans receive larger attention from the integrated platform. I also show that the right tail of the quality distribution has important implications for the optimal allocation of producers. In particular, as the right tail of the quality distribution shrinks, product diversity increases.

My final main result shows that even when production is decentralized, there is scope in these industries for a recommendation system to improve upon the competitive equilibrium. A recommendation system connects consumers to products, but unlike the integrated platform is unable to direct production. Instead, the recommendation system influences producers by selectively *suppressing* content. By suppressing mediocre content in high-demand genres, the recommendation system discourages producers from over-producing in these genres. This encourages producers to produce a wider range of content than they would in the competitive equilibrium.

## 1.1 Related Literature

There exists a mainly empirical literature studying how digitization has affected cultural industries. A major theme in this literature is the trade-off between digitization leading to increased piracy and decreased profits, and how technological developments have decreased the cost of entry. Waldfogel (2013) studies exactly these two completing forces in the case of the movie industry, and Aguiar and Waldfogel (2016) studies these forces in the music industry. Piolatto and Schuett (2012) studies how

---

<sup>3</sup>This result goes against standard arguments where allowing firms to internalize the value of their production leads to efficiency.

piracy disproportionately hurts small producers. Most related to my study in spirit is Aguiar and Waldfogel (2018), which evidences quality uncertainty in the production of music, and shows that this uncertainty has large welfare implications. There is also a small literature studying how streaming services have influenced the music industry. Gambato and Sandrini (2025a) and Gambato and Sandrini (2025b) study producer incentives in streaming services, including the impact of promotional services. Aguiar and Waldfogel (2021) provides evidence of significant power of streaming services to promote certain products, and Aguiar, Waldfogel, and Waldfogel (2021) measures bias in Spotify promotion of new music.

My paper also contributes to the growing literature on the economic impact of recommendation systems. Most work studying the impact of these intermediary platforms has focused on how these platforms leverage informational asymmetries to influence consumer behaviour. Empirically, Lee and Musolff (2025) studies the impact of Amazon’s Buybox on consumption and consumer surplus. Calvano et al. (2025) studies how algorithmic recommendations affect the distribution of consumption choices and market concentration, but do not focus on producer incentives. Most related is Chen, Li, and Preuss (2025), which studies the scope of recommendation systems to influence producer effort on social media platforms. While I also study the ability of recommendation systems to influence producer behaviour, I focus on producer sorting across horizontally differentiation, whereas Chen, Li, and Preuss (2025) assumes that producers are exogenously anchored to a product type in order to focus on effort choices.

At a technical level my paper speaks to the contest literature with noisy output. Since producers are uncertain about their quality ex-ante in my model, genre decisions are related to sorting across one of multiple contests. Drugov and Ryvkin (2020) and Ryvkin and Drugov (2020) study entrant effort in single noisy contests. Grossmann (2021) and Deng et al. (2025) study concurrent contests with deterministic output. While at a high-level my model is related to this literature, my focus is different. I am not interested in the design of contests, since “prizes” correspond to consumer demand, which is not a design choice. Instead, I am interested in the implications of noisy output on producer sorting across contests, with an exogenous prize structure.

**Outline of Paper.** Section 2 introduces a motivating example to outline the forces

in the main model. [Section 3](#) develops the full model, and [Section 4](#) performs some preliminary analysis. The main results of the paper are presented in [Section 5](#), and comparative statics are developed in [Section 6](#). Finally, the recommendation system is introduced in [Section 7](#) and [Section 8](#) concludes.

## 2 Motivating Example

In this section I present an extremely stylized example in order to fix ideas.

**Producers and Products.** There are two producers and two product types, which I refer to as “genres”. Producers are ex-ante identical, but there is uncertainty over their production technology. Producers’ quality  $v$  is distributed according to the uniform distribution on  $[a, b]$  ( $a, b \in \mathbb{R}_+$ ), and is independent across producers. Importantly, producers observe their production technology only *after* they commit to a genre: they do not observe their quality until after production.

**Consumers.** Demand differs for the two genres. Mass  $\mu > \frac{1}{2}$  of consumers consume content in genre 1, and mass  $1 - \mu$  consume content in genre 2. Within each genre, each consumer can consume at most one product, and receives utility from consumption that is equal to their consumed product’s quality.

**Prices.** Finally, suppose that producers are price takers in these markets. In this example, I consider two different pricing schemes. In the *flat-pricing* scheme, consumers pay a fixed-price of  $p \leq a$  for whichever good they purchase. In the *variable-pricing* scheme, consumers pay  $v$  for a good of quality  $v$ .

I now present the main result from this motivating example: uncertainty over their own quality results in producers concentrating too heavily in one genre.

### Proposition 1.

- (i) If  $\mu > \frac{2}{3}$ , there is a unique equilibrium of the fixed-price scheme where both producers produce in genre 1.
- (ii) If  $\mu > \frac{3a+3b}{4a+5b} \in [\frac{3}{5}, \frac{2}{3}]$ , then there is a unique equilibrium of the variable-price scheme where both producers produce in genre 1.
- (iii) If  $\mu < \frac{3}{2} \cdot \frac{a+b}{a+2b} \in [\frac{3}{4}, \frac{9}{10}]$ , both producers producing in genre 1 is strictly inefficient.

If demand is concentrated in genre 1, but not too concentrated, the unique equilibrium of this game is inefficient regardless of which pricing scheme is used. The intuition for the possible inefficiency is as follows. When a producer is uncertain over their own quality, they aim to maximize their own expected profits. In doing so, firms do not internalize the externality that comes from their displacement of other products. Despite its simplicity, this simple example highlights several key forces that will be present in the main model.

**The effect of ex-ante uncertainty.** If producers knew their quality before making production decisions, the inefficiency would vanish. The high-quality producer will claim the larger demand in genre 1, and the low-quality firm will then produce in genre 2, knowing that it cannot compete with the better firm in genre 1.

**The effect of pricing.** The range of  $\mu$  for which there is an inefficiency grows when prices reflect the underlying quality of the good produced, which goes against standard intuition. If producers converge on producing in the same genre, conditional on being consumed, their realized value is expected to be higher than in the smaller genre. This creates an additional force that drives the two producers to produce in the same genre. While these are two extreme pricing examples, they broadly capture forces in the market for cultural products. Flat-pricing schemes have been widely documented in these industries,<sup>4</sup> but it is hypothetically possible for producers to price goods in a way that depends on their underlying quality. I do not allow producers to undercut one another on price to reflect the historical reality that prices for most cultural products were set by intermediaries (e.g., movie theatres, record stores). Hence, it was simply not possible for a producer of a movie to offer a cheaper movie ticket in order to attract customers.

**The effect of the distribution of quality.** Third, this inefficiency is most pronounced when  $a$  is close in value to  $b$ . That is, the inefficiency is most pronounced when there is less dispersion in the distribution of quality. In this case, the uncertainty that firms face in terms of profits is due not to the cardinal value of the good they produce, but about their *ordinal* rank across other producers. In the limiting case when  $a \approx b$ , a planner who assigns producers across genres does not care about

---

<sup>4</sup>See Hennig-Thurau, Houston, et al. (2019) for instance.

realized quality, and so will always distribute the two across the different genres.

**Scope for an intermediary.** Finally, consider a possible intervention from an intermediary who is able to artificially suppress content. This intermediary can commit to rejecting a product from the market, based on the product’s genre and quality. In the flat pricing scheme, if  $\mu \gtrsim \frac{2}{3}$ , by rejecting content in genre 1 below some threshold  $\tau \geq a$ , the platform decreases expected profits from genre 1. If  $\tau$  is sufficiently large, this risk of suppression encourages producers to diversify across the two genres. Whether this increases consumer utility is ambiguous. There is a welfare loss relative to the first-best because realizations of quality in  $[a, \tau]$  for the producer who produces in genre 1 will be rejected. However, there are also welfare gains if the original concentrated equilibrium was inefficient. As long as  $\tau$  is not too large, this second force will dominate and it will be welfare-improving to suppress content. This highlights the potential scope for an intermediary to improve welfare without directly controlling producers.

Although in this simple example allocative inefficiency exists only for specific values of  $\mu$ , in the general model allocative inefficiency generically persists. The general model additionally allows me to demonstrate in a more nuanced way the ways in which market structure, consumer heterogeneity, and uncertainty over quality influence allocative inefficiency.

### 3 Model

I now describe the general framework that I will use to build upon the analysis of the motivating example.

**Producers’ Choice of Genre and Quality Realization.** There is a fixed mass  $m$  of ex-ante identical producers. These producers each produce one good, which is differentiated both along a horizontal “genre” dimension, and a vertical “quality” dimension. As in the motivating example, while producers are able to choose their genre, their quality is random, and is realized *after* they make their choice over genre.

There are  $G \in \mathbb{N}$  genres. I abuse notation and also use  $G = \{1, \dots, G\}$  to denote the set of genres. A producer’s choice consists of which genre  $g \in G$  in which to produce. After this decision, the producer’s quality  $v$  is realized according to a continuous distribution  $F$ , independent across genres. Let  $\bar{F}(v) = 1 - F(v)$  denote the



survival function of the distribution, and  $f$  its density. Unless otherwise indicated, I assume that  $F$  has increasing hazard rates:

**Assumption 1.**  $F$  has increasing hazard rates:  $\frac{f(v)}{F(v)}$  is strictly increasing in  $v$  on its domain.

Each producer's quality is realized independently. Denote by  $\Psi : [0, m] \rightarrow \Delta G$  a population strategy for producers, where  $\Psi(i) \in \Delta G$  denotes the strategy of producer  $i$ . Then, define  $\psi_g := \int_0^m \Psi_g(i) di$  as the expected mass of producers in genre  $G$ . Since producers are small, the measure  $\psi \in m\Delta G$  implies a deterministic measure of realized products  $\mu_g^\psi$  in each genre  $g \in G$ . In particular, this measure is given by  $\mu_g^\psi((-\infty, v]) = \psi_g F(v)$  for all  $v \in \mathbb{R}$ .

**Consumers.** There is a unit mass of consumers. A consumer is defined by a two dimensional type: the genre which he consumes, and the maximum quantity of content that he is able to consume (his “time” constraint). A consumer type is hence a tuple  $(g, t) \in \Theta := G \times \mathbb{R}_+$ .

Let  $T \in \Delta \Theta$  denote the distribution of consumer types. From this, one can derive the distribution  $T_g$  of demand for genre  $g$ . Here,  $T_g(t)$  denotes the mass of consumers who are willing to consume at most  $t$  units of content from genre  $g$ :

$$T_g(t) = T\left(\{(g', t') | g' \neq g, \text{ or } g = g', t' \leq t\}\right), \quad \text{for all } t \geq 0$$

In particular,  $T_g(0) = T(G \setminus \{g\} \times \mathbb{R}_+)$  is the mass of consumers who consume no content of genre  $g$ . As for the distribution over quality, let  $\bar{T}_g(t) = 1 - T_g(t)$  denote the survival function of consumer demands for genre  $g$ . It will be useful to define  $c_g := \bar{T}_g(0)$ , the possible consumers of genre  $g$ . This is the total mass of consumers who demand a non-zero amount of content of type  $g$ . I place the following assumption on the distribution of consumer types. It is standard, and weaker than [Assumption 1](#).

**Assumption 2.**  $T_g$  is *regular*. That is,

$$t - \frac{\bar{T}_g(t)}{dT_g(t)}$$

is weakly increasing in  $t$  on the support of  $T_g$ , which is  $[0, \bar{t}_g]$  for some  $\bar{t}_g$ .

**Consumer Preferences.** The utility that a consumer of type  $(g, t)$  receives from consuming a good from genre  $g$  is proportional to its quality: he receives  $v \cdot dv$  from consuming a good of quality  $v$ . If the marginal price of a good is zero, the consumer would hence first consume the highest-quality goods, and only stop once he reaches his time constraint  $t$  or there are no goods of positive quality remaining.

Explicitly, let  $\mu_g$  denote the measure of content in genre  $g$ . To find the value at which his time constraint binds, first solve for  $v_t$  such that

$$\mu_g([v_t, \infty)) = t$$

If no such  $v_t$  satisfies the equation, define it as  $-\infty$ . The consumer will hence consume all goods in genre  $g$  that are of quality above  $\max\{0, v_t\}$ . Since his utility is proportional to the quality of the goods that he consumes, his utility is then

$$\int_{\max\{0, v_t\}}^{\infty} v d\mu_g(v)$$

In practice,  $\mu_g = \mu_g^\psi$  takes the form described above. In this case,  $v_t$  will solve

$$\psi_g \bar{F}(v_t) = t$$

This implies that the consumer of type  $(g, t)$  will consume all content whose value is weakly greater than  $\max\left\{0, \bar{F}^{-1}\left(\frac{t}{\psi_g}\right)\right\}$ . Defining

$$W(x) := \int_{\bar{F}^{-1}(x)}^{\infty} v f(v) dv$$

the consumer's utility from consumption is hence

$$\psi_g W\left(\min\left\{\bar{F}(0), \frac{t}{\psi_g}\right\}\right).$$

**Prices and Producer Profits.** For now, suppose that all goods, regardless of value, command the same price  $p$  in the market.<sup>5</sup> This reflects the well-documented reality of flat pricing in many entertainment markets: there is extremely little price

---

<sup>5</sup>I relax this assumption later.

variation across movie tickets or albums.<sup>6</sup> In this case, the consumer will stop consumption either at the value  $v$  where his time constraint is reached, or at the baseline price  $p$ .

His utility from consumption can be written as

$$\underbrace{\psi_g W\left(\min\left\{\bar{F}(p), \frac{t}{\psi_g}\right\}\right)}_{\text{Utility from Consumption}} - \underbrace{\psi_g p \cdot \min\left\{\bar{F}(p), \frac{t}{\psi_g}\right\}}_{\text{Cost of Consumption}}$$

Producers anticipate demand in each genre, and enter whichever genre maximizes their expected profits. Given strategies for other producers (and hence,  $\psi_g$ ), expected profits from an individual consumer of type  $(g, t)$  are given by

$$p \cdot \min\left\{\bar{F}(p), \frac{t}{\psi_g}\right\}$$

Hence, expected profits for a given producer in genre  $g$  are given by

$$p \cdot \int_0^\infty \min\left\{\bar{F}(p), \frac{t}{\psi_g}\right\} dT_g(t)$$

Henceforth I use the convention that  $\frac{0}{0} = 0$ , so that there is continuity in this expression at  $\psi_g = 0$ : expected profits with zero competition is  $p\bar{F}(p) \cdot c_g$ .

**Timing.** The timing of the game is as follows:

1. Producers commit to genres, inducing  $\psi$ .
2. Individual producer quality is realized.
3. Consumers purchase content to maximize their utility.
4. Producer profits are realized.

## 4 Preliminaries: Competitive Equilibria and the Integrated Platform

In this section I solve for the unique competitive equilibrium given a price  $p$ . I also introduce the integrated platform, and solve for its optimal allocation of producers.

---

<sup>6</sup>See Hennig-Thurau, Houston, et al. (2019) for a discussion.

#### 4.1 Competitive Equilibria

As noted above, facing competition  $\psi \in m\Delta G$ , a producer's expected profits from entering genre  $g$  are given by

$$p \cdot \int_0^\infty \min \left\{ \bar{F}(p), \frac{t}{\psi_g} \right\} dT_g(t)$$

If  $\psi_g = 0$ , then producers face no competition in genre  $g$ . This means that all consumers of type  $(g, t)$  will be willing to consume their product, for any  $t$ . Their product will be consumed by all agents if its realized quality is above the posted price  $p$ . Expected profits in this case are hence  $p\bar{F}(p)c_g$ . If  $\psi_g > 0$ , then consumers are constrained and become picky about which goods they consume. The probability that an individual producer's good is consumed by such a consumer is now  $\frac{t}{\psi_g}$ , since a product must be in the upper  $t$  mass of content to be consumed. As  $\psi_g$  becomes large, other genres become relatively attractive, and producers will shift away from genre  $g$ .

A competitive equilibrium, then, is an allocation of producers such that each producer is operating in a genre that maximizes its expected profits.

**Definition 1.** A competitive equilibrium at price  $p \geq 0$  is an allocation of producers  $\psi \in m\Delta G$  such that

$$\psi_g > 0 \iff g \in \operatorname{argmax} \int_0^\infty \min \left\{ \bar{F}(p), \frac{t}{\psi_g} \right\} dT_g(t)$$

If  $\psi_g > 0$ , then some mass of producers is operating in genre  $g$ . Hence, it must be that genre  $g$  maximizes expected profits across all genres. This does not necessarily imply that profits are the same across all genres. If demand is sufficiently low in a genre, then it may not be sufficiently attractive to producers even in the face of zero competition, so that  $\psi_g = 0$ .

Since  $p$  is constant, I drop the leading  $p$  from the definition in order to extend the definition to a price of  $p = 0$ . This setting represents those in which producers benefit from consumption, but consumers do not directly pay producers (e.g., through advertising revenue). Given that profits in any genre  $g$  are strictly decreasing in  $\psi_g$ , there is at most one competitive equilibrium.

**Lemma 1.** *For any  $p \geq 0$ , a unique competitive equilibrium allocation exists. Denote this allocation of producers  $\psi^p \in m\Delta G$ .*

Of importance is the set of genres that receive some producers in the unique competitive equilibrium. I call these genres “active” in that some content is being produced in these genres.

**Definition 2.** Given a price  $p \geq 0$ , define the set of *active* genres in the competitive equilibrium at price  $p$  as  $A^p := \{g \in G | \psi_g^p > 0\}$ .

## 4.2 Integrated Platform

I now turn to defining and solving the integrated platform’s problem. The market structure with an integrated platform differs from the competitive equilibrium. First, being integrated, the platform operates all producers. This means that it decides  $\psi$  directly, unlike in the competitive equilibrium where  $\psi$  is determined by the decentralised choices of producers. Second, since the platform sells its own goods, it is not beholden to a fixed market price. This means that the platform can sell individual bundles of goods to consumers. The timing of the game hence differs slightly:

1. The platform decides on an allocation  $\psi$  of producers across genres.
2. The platform observes realizations of quality, and creates and prices bundles of goods.
3. Consumers choose their bundle to maximize expected utility.
4. Platform profits are realized.

These two changes allow the platform to extract full surplus from consumers. To see this, after the decision of  $\psi$ , facing no price, a consumer of type  $(g, t)$  is willing to consume all content above the threshold

$$\max \left\{ 0, \bar{F}^{-1} \left( \frac{t}{\psi_g} \right) \right\}$$

yielding utility

$$\psi_g W \left( \min \left\{ \bar{F}(0), \frac{t}{\psi_g} \right\} \right)$$

The heterogeneity in consumer preferences comes only through the extensive margin of *how much* of the good to consume. Hence, the platform can create a family of bundles, one for each  $(g, t)$  at price  $\psi_g W\left(\min\left\{\bar{F}(0), \frac{t}{\psi_g}\right\}\right)$ , to extract all of the consumer's surplus.

**Definition 3.** An integrated-optimal allocation of producers  $\psi^I \in m\Delta G$  is one which maximizes

$$\sum_{g \in G} \int_0^\infty \psi_g W\left(\min\left\{\bar{F}(0), \frac{t}{\psi_g}\right\}\right) dT_g(t)$$

I now solve the platform's problem, and then turn to comparing the platform's optimal allocation of producers to the competitive equilibrium allocation of producers.

#### 4.2.1 Solving the Platform's Problem

In order to solve for the optimal allocation of producers  $\psi^I$ , consider first the effect of increasing the mass of producers  $\psi_g$  in a given genre  $g$ . More specifically, consider how a consumer of type  $(g, t)$ 's utility increases in  $\psi_g$ . There are two cases to consider.

**Case 1:**  $t > \psi_g \bar{F}(0)$ . Here, the consumer's time constraint does not bind. Hence, he is willing to consume any new content whose realized value is positive so that his marginal increase in utility is equal to

$$\int_0^\infty v f(v) dv$$

**Case 2:**  $t \leq \psi_g \bar{F}(0)$ . In this case, the consumer's time constraint does bind. There are two differences between this case and case 1. First, the consumer only consumes new content whose realization is above his threshold of consumption  $\bar{F}^{-1}\left(\frac{t}{\psi_g}\right)$ . Second, if he does consume this new content, he must replace existing consumption. He does so by first stopping to consume the content that is exactly at his threshold; content of value  $\bar{F}^{-1}\left(\frac{t}{\psi_g}\right)$ . Hence, his increase in utility is given by

$$\int_{\bar{F}^{-1}\left(\frac{t}{\psi_g}\right)}^\infty \left(v - \bar{F}^{-1}\left(\frac{t}{\psi_g}\right)\right) f(v) dv$$

Similar to the definition of  $W(x)$ , define

$$M(x) := \int_{\bar{F}^{-1}(x)}^{\infty} \left( v - \bar{F}^{-1}(x) \right) f(v) dv$$

Putting these together, we have the following result.

**Lemma 2.** *For all  $\psi_g \geq 0$ ,  $\psi_g W\left(\min\left\{\bar{F}(0), \frac{t}{\psi_g}\right\}\right)$  is differentiable with*

$$\frac{\partial}{\partial \psi_g} \left[ \psi_g W\left(\min\left\{\bar{F}(0), \frac{t}{\psi_g}\right\}\right) \right] = M\left(\min\left\{\bar{F}(0), \frac{t}{\psi_g}\right\}\right)$$

Moreover  $M$  is strictly increasing and strictly convex on  $[0, \bar{F}(0)]$ ,  $W$  is strictly increasing and strictly concave on  $[0, \bar{F}(0)]$ , with  $M(0) = W(0) = 0$ , and  $M(\bar{F}(0)) = W(\bar{F}(0))$ .

It can be seen immediately that since  $M$  is increasing,  $\psi_g W\left(\min\left\{\bar{F}(0), \frac{t}{\psi_g}\right\}\right)$  is weakly concave in  $\psi_g$ . Hence, it is possible to use this marginal increase functions  $M$  in order to derive the integrated platform's optimal allocation.

**Proposition 2.** *The integrated optimal allocation of producers  $\psi^I$  is unique. Moreover, it is the allocation of producers such that*

$$\psi_g^I > 0 \iff g \in \operatorname{argmax} \int_0^\infty M\left(\min\left\{\bar{F}(0), \frac{t}{\psi_g^I}\right\}\right) dT_g(t).$$

Importantly, the condition for optimality given in [Proposition 2](#) is appears quite similar to the condition for a competitive equilibrium ([Definition 1](#)). The interpretation is different since producers are assigned to different genres by the centralized platform. However, the intuition is the same. If demand in a genre is too low, then the platform may choose to not invest any producers in producing content for that genre. This is true even though the monopolist is able to extract all value from goods produced in that genre.

As for the competitive equilibrium, the set of active genres in the integrated platform's optimal allocation will prove important.

**Definition 4.** Define the set of active genres in the integrated optimal allocation  $A^I := \{g \in G \mid \psi_g^I > 0\}$ .

## 5 Market Failures

Since the integrated platform maximizes total surplus, deviations from  $\psi^I$  represent losses in surplus. In this section I describe the main result of the paper: ex-ante uncertainty over quality leads to deviations from  $\psi^I$  in all competitive equilibria. I also show that flat pricing minimizes this inefficiency: allowing producers to absorb more of the social benefit of their products worsens the inefficiency.

It is important to mention that there exists a different market failure that the platform is able to overcome. Due to ex-ante uncertainty over quality, some goods have realized values  $v \in (0, p)$ . These goods are not consumed in the competitive equilibrium with price  $p$  since the price is deterministic, even though there may be some consumers who want to consume these goods. The platform overcomes this failure by setting its own prices: it is able to charge low prices for low-quality goods. This “posted-price problem” is an important failure of the market, but doesn’t rely on the specifics of the setting. Any market with vertical differentiation and prices set by an intermediary is subject to the same issue.

The existence of one market failure does not preclude the possibility of a different market failure, however. An important feature of these markets is that producers sort across different product types. Moreover, this sorting depends importantly on the market structure. In competitive equilibria, sorting is decentralized: producers sort in order to maximize individual expected profits. When the market is integrated, sorting is organized by the platform and has been shown to be efficient. In this section I highlight the existence of “allocative” inefficiency: inefficiency in producer sorting across genres. In particular, I show that allocative inefficiency exists generically whenever there is ex-ante uncertainty over quality.

**Definition 5.** I say that there is *allocative inefficiency* in allocation  $\psi$  if  $\psi \neq \psi^I$ .

### 5.1 Ex-ante Quality Realization

In order to understand why ex-ante uncertainty in quality is intertwined with allocative inefficiency, I first describe an alternative setting where producers’ quality is realized before they decide their genre. As I show, echoing the motivating example in [Section 2](#), allocative inefficiency disappears if producers know their quality before sorting across genres.



In this setting, there is a mass  $m$  of producers, of differing quality. Producers' quality is distributed according to distribution  $F$ , but now producers see their quality before deciding in which genre to produce.

Strategies for producers can now be written as  $\Psi : \mathbb{R} \rightarrow \Delta G$ , where  $\Psi(v) \in \Delta G$  represents the strategy for producers of realized quality  $v$ . A strategy induces a measure  $\mu_g$  of quality in each genre. Notice that, in this setting, the mass of producers in a genre is no longer a sufficient statistic for  $\mu_g$ . In the competitive equilibrium, producers still aim to maximize profits given their competitors' choice of horizontal differentiation. In the integrated platform's optimal allocation, the platform assigns producers to different genres in order to maximize total surplus.

In this setting where producers know their quality before they enter a genre, the issue of a posted price persists: the platform is able to increase the total surplus by providing low-value goods at low cost. However, as the following result shows, this is the only market failure in this setting: producers whose products are consumed sort efficiently.

**Proposition 3.**  $\mu^I|_{[p,\infty)}$  and  $\mu^P|_{[p,\infty)}$  are unique, for all  $p \geq 0$ . Moreover,  $\mu^I|_{[p,\infty)} \equiv \mu^P|_{[p,\infty)}$ , for all  $p \geq 0$ .

The intuition for this result is straightforward. Consider the choice of the “best” producer.<sup>7</sup> Facing no competition, this producer will go to the genre with the highest demand. The platform also sorts this producer to that same genre, because welfare is supermodular in value and demand. The “next-best” producer, considering the best producer's choice as given, update its expected demand, and sorts in the same way. This positive-assortative sorting maximizes each individual producer's profits, and also maximizes the platform's overall profits.

## 5.2 Allocative Inefficiency

When quality is realized ex-ante, producers' and the platform's incentives are aligned: they both aim to maximize the mass of consumers that consume high-quality goods. As the motivating example shows, this alignment can break-down when producers are unaware of their own quality. This observation extends to the full model: there is generically allocative inefficiency.

---

<sup>7</sup>Of course, since  $F$  represents a continuous distribution, there is no “best” producer. However, this language is useful for conveying the intuition behind the result.

**Theorem 1.** *Fix  $p \geq 0$ . If  $|A^I| > 1$ , generically  $\psi^p \neq \psi^I$ .*

Before I explain the intuition underlying the result, several comments on the result are needed. First, the requirement that  $|A^I| > 1$  is necessary to rule out degenerate cases where  $\psi^I = m\delta_g$  for some  $g \in G$ . As I show below,  $|A^I| = 1$  implies  $A^p = A^I$  for all  $p \geq 0$ , and so the two agree. Second, genericity is with respect to the distribution of type  $T$ . It is necessary to rule out edge-cases such as  $T_g = T_{g'}$  for all  $g, g'$  (where  $\psi_g^I = \psi_g^p = \frac{m}{G}$  for all  $g, p$ ).

The intuition for the result is straightforward. As we have seen, in the competitive equilibrium

$$\int_0^\infty \min \left\{ \bar{F}(p), \frac{t}{\psi_g} \right\} dT_g(t)$$

is equal across all active genres. In the platform's optimal allocation, on the other hand,

$$\int_0^\infty M \left( \min \left\{ \bar{F}(0), \frac{t}{\psi_g} \right\} \right) dT_g(t)$$

is equal for all active genres. By [Lemma 2](#),  $M$  is strictly convex (and in particular, not linear). Since the distributions  $T_g$  have no ex-ante relationship, there is no reason why equality across one expectation should imply equality for the other expectation.

[Proposition 2](#) and [Theorem 1](#) together highlight the existence of a fundamental difference between settings in which producer quality is realized before production decisions are made, and those in which producer quality is realized ex-post. In the former case, efficient sorting of producers is robust regardless of the prevailing market price. On the other hand, as soon as producers face uncertainty over their quality, they sort inefficiently, regardless of the market price. In the next section I show how this inefficiency manifests, and varies with the market price.

### 5.2.1 The Form of the Failure

[Theorem 1](#) highlights the robust existence of allocative inefficiency in markets where producers are unaware of their own quality at the time of their production choice. However, it provides no insight into what form the market failure takes. In this

section I show that the theory admits consistent predictions in terms of how this failure manifests. In particular, allocative inefficiency leads to a smaller range of products in competitive equilibrium than is socially optimal.

**Definition 6.** I say that market structure  $S$  features *less range* than  $S'$  if  $A^S \subseteq A^{S'}$ .

This ordering of market structures is incomplete: if  $A^S$  and  $A^{S'}$  are not nested then they cannot be compared. Importantly, the strictness of this definition captures exclusively settings where moving from market structure  $S$  to market structure  $S'$  leads to a widening of produced content, without losing any genres.

**Theorem 2.** For all  $F$  and  $T$  satisfying [Assumption 1](#) and [Assumption 2](#),

- (i) If  $0 \leq p < p'$  the competitive equilibrium with price  $p'$  features less range than that with price  $p$ :  $A^{p'} \subseteq A^p$ ;
- (ii) All competitive equilibria have less range than the platform's optimal allocation: for all  $p \geq 0$ ,  $A^p \subseteq A^I$ .

*Each inclusion may be strict.*

This result highlights the specific ways in which producers deviate from the socially efficient allocation of production. I first describe the intuition, and then give an outline of the proof of the result.

When individual producers choose genres, they maximize expected profits. They do not internalize the effect that a high-value realization of their quality has on lower-value producers. If producer quality is known, then these low-value producers will respond by moving to a less-competitive genre. When quality is not known ex-ante, producers are hence driven to compete too much in genres where demand is high, because the profits following a realization of high quality is very large.

The platform has two benefits over these producers. First, it is able to hedge against uncertainty since it controls the entire mass of producers. Second, and more importantly, it internalizes the impact of this displacement through  $M$ . When a genre is concentrated, the platform benefits little from increasing production in that genre, since it leads to displacement of already high-quality goods. This leads the platform to produce a wider range of content than individual producers in competitive equilibrium.

**Proof Outline.** Regardless of the market structure, genres are added in order of  $c_g$ . This means that in all cases comparing range is possible (i.e., there will be nesting). To see (i), expected profits at  $\psi^p$  are given by (after normalizing by  $p$ )

$$\int_0^\infty \min \left\{ \bar{F}(p), \frac{t}{\psi_g} \right\} dT_g(t) = \bar{F}(p) \bar{T}_g(\psi_g \bar{F}(p)) + \int_0^{\psi_g \bar{F}(p)} \frac{t}{\psi_g} dT_g(t)$$

A genre  $g'$  is active if and only if

$$\bar{F}(p) \cdot c_{g'} > \bar{F}(p) \bar{T}_g(\psi_g \bar{F}(p)) + \int_0^{\psi_g \bar{F}(p)} \frac{t}{\psi_g} dT_g(t)$$

for some (equivalently, all) active  $g$ . We can rewrite this as

$$c_{g'} > \bar{T}_g(\psi_g \bar{F}(p)) + \frac{1}{\bar{F}(p)} \int_0^{\psi_g \bar{F}(p)} \frac{t}{\psi_g} dT_g(t)$$

While the left-hand side is constant in  $p$ , it can be shown that the right-hand side is increasing in  $p$ . Hence, it becomes harder for this condition to be satisfied as  $p$  increases. The intuition is straightforward: as the price  $p$  increases, it becomes harder for producers to be “above the bar”. This decreases relative concentration in high-demand genres since fewer of my competitors will be above the bar. At the same level of production in a genre, then, the genre is relatively more attractive. This decreases the congestion effect that pushes producers to explore new genres.

By (i), it is sufficient to check (ii) in the case of  $p = 0$ . At an allocation  $\psi$ , a genre would become active in the  $p = 0$  competitive equilibrium if and only if

$$\bar{F}(0) c_{g'} > \int_0^\infty \min \left\{ \bar{F}(0), \frac{t}{\psi_g} \right\} dT_g(t)$$

In the integrated optimal allocation, a genre is active if and only if

$$M(\bar{F}(0)) \cdot c_{g'} > \int_0^\infty M \left( \min \left\{ \bar{F}(0), \frac{t}{\psi_g} \right\} \right) dT_g(t)$$

Now,  $\min \left\{ \bar{F}(0), \frac{t}{\psi_g} \right\} dT_g(t)$  represents the distribution of a random variable distributed on the interval  $[0, \bar{F}(0)]$ . Similarly,  $c_{g'} \circ \bar{F}(0) + (1 - c_{g'}) \circ 0$  is also a random

variable distributed on the same interval. If the first condition holds, then the mean of the first random variable is smaller than the mean of the second. Since  $M$  is strictly convex with  $M(0)$ , this implies that if the first condition holds then the second condition holds.

### 5.3 Quality-Dependent Pricing

[Theorems 1](#) and [2](#) show that when prices are fixed and flat, ex-ante uncertainty over producer quality leads to inefficiencies in producer sorting. It is natural to ask what role flat pricing plays in driving this market inefficiency. I motivated a market with inflexible flat pricing by its well-documented presence in these industries.<sup>8</sup> Intermediaries (e.g., record stores and movie theatres) set prices, and do so without necessarily considering the effects these prices have on producer incentives. Intuition suggests that if firms internalized the value of their product through price discrimination, allocative inefficiencies would disappear.

I show in this section that the opposite occurs. When producers charge higher prices for higher-value goods, the degree of allocative inefficiency *increases*. The reasoning is similar to that seen in the motivating example. [Theorem 2](#) shows that producers are too concentrated in equilibrium. This means that they over-serve high-demand genres, so that consumers in these genres consume content that is of higher value than is socially optimal. When prices depend on realized quality, producers place *more* weight on these consumers: conditional on the producer’s product being consumed by these consumers, it must be of high value, so that the product commands a large price. On the other hand, under-served genres are attractive in the competitive equilibrium because there is little competition. But little competition means that products are consumed even if they are of low-value and hence command a low price. I now formalize this argument.

**Definition 7.** A pricing function  $P : \mathbb{R} \rightarrow \mathbb{R}_+$  is called *admissible* if  $P$  is differentiable with  $P'(v) \in [0, 1]$  for all  $v$ .

Admissibility ensures that consumers will purchase higher-quality goods when given the possibility. While in many markets it is common for higher-quality goods to be priced so as to dissuade consumption by some consumers, this is uncommon

---

<sup>8</sup>See Hennig-Thurau, Houston, et al. ([2019](#)).

in the markets under study. As noted, pricing in these industries is relatively flat. Admissible pricing functions represent a deviation from flat pricing that maintains the spirit of consumers consuming the best content that is available to them.

Admissibility ensures that there exists a minimum quality  $\underline{v}_P \in \mathbb{R}_+$  such that  $v \geq P(v)$  if and only if  $v \geq \underline{v}_P$ . Call this value the “base price” of pricing function  $P$ . With this language it is possible to reinterpret the pricing function as containing a minimum price  $\underline{v}_P$ , and then some additional quality surcharge  $P(v) - \underline{v}_P$  above this price.

**Proposition 4.** *For any admissible pricing function  $P$ , there exists a unique competitive equilibrium allocation of producers  $\psi^P$ . Given two admissible pricing functions  $P_1, P_2$  with  $\underline{v} := \underline{v}_{P_1} = \underline{v}_{P_2}$ . If  $\frac{P_2(v)}{P_1(v)}$  is increasing on  $[\underline{v}, \infty)$ , then the competitive equilibrium under pricing function  $P_2$  features less range than that under  $P_1$ :  $A^{P_2} \subseteq A^{P_1}$ .*

Proposition 4 shows that increasing the quality surcharge decreases the range of products that are produced in the equilibrium. Hence, the more that firms are able to extract the value of their good, the more the competitive equilibrium is pushed away from the socially optimal allocation. An important corollary of Proposition 4 is that adding any price differentiation to the flat-price competitive equilibrium will decrease the range of products that are offered.

**Corollary 1.** *For any admissible  $P$ , the competitive equilibrium under  $P$  features less range than the flat-price equilibrium at  $P$ ’s base price:  $A^P \subseteq A^{\underline{v}_P}$ . Hence,  $A^P \subseteq A^{\underline{v}_P} \subseteq A^0 \subseteq A^I$  for all admissible  $P$ .*

In this section I have showed the prevalence of allocative inefficiency when producers are uncertain over their own quality. In the following section I perform comparative statics on the primitives of the model in order to show how these primitives influence the allocation of producers across genres.

## 6 Comparative Statics

In this section I present comparative statics that aim to enrich the results of Theorems 1 and 2. I first perform comparative statics on the distribution of consumer preferences  $T_g$ . I then perform comparative statics on the distribution of quality realizations  $F$ .

### 6.1 The effect of “super-fans.”

An important dimension of difference across genres is the degree of heterogeneity within a genre. Consider two genres  $g, g'$  with the same total demand:  $\mathbb{E}_{T_g}[t] = \mathbb{E}_{T_{g'}}[t]$ . If  $T_{g'}$  is a mean-preserving spread of  $T_g$ , the population of consumers who consume content of genre  $g$  is more heterogeneous than those who consume content from genre  $g'$ . In particular, there will be more “super-fans” in genre  $g'$ —consumers who consume a large amount of content—and also more consumers who consume only a small amount, or no, content.

In the competitive equilibrium, increasing heterogeneity in consumers through a mean-preserving spread results in some consumers consuming more and other consumers consuming less. The increased consumption is now of lower-value goods than the demand that is lost, however. This suggests that more producers will produce in genre  $g$  than in genre  $g'$ : super-fans dampen the attractiveness of a genre.

For the platform, the story is more nuanced. Due to the concavity of  $W$ , a more heterogeneous population of consumers means that at the same level of production  $\psi_g = \psi_{g'}$ , genre  $g'$  will produce lower profits than genre  $g$ . When the mass of producers is small, this suggests that, analogous to the competitive equilibrium, there are more producers assigned to genre  $g$ . However, when the mass of producers is large, the result flips, and  $g'$  receives more attention. The reasoning is as follows: high-demand consumers necessarily consume low-value goods. Increasing the mass of producers in a genre benefits these high-demand consumers more than it benefits low-value consumers, both because they consume more of the new content (because of their higher overall demand) and because they replace worse content. Convexity of  $M$  implies that this increase in benefit is increasing in the demand of consumers, so that by spreading out the mass of consumers, the genre becomes more attractive to the platform. These results are summarized in the following proposition.

**Proposition 5.** *Consider two genres  $g, g'$  with  $T_{g'}$  a mean-preserving spread of  $T_g$ . There exists a mass of producers  $\bar{m} \in \mathbb{R}$  such that*

- (i) *More producers produce in genre  $g$  than genre  $g'$  in every competitive equilibrium:  $\psi_g^P \geq \psi_{g'}^P$  for all admissible  $P$ .*
- (ii) *When the mass of producers is small, the platform assigns more producers to  $g$*

than  $g'$ : for all  $m \leq \bar{m}$ ,  $\psi_g^I \geq \psi_{g'}^I$ ; but

(iii) when the mass of producers is large, the platform assigns more producers to  $g'$  than  $g$ : for all  $m > \bar{m}$ ,  $\psi_g^I < \psi_{g'}^I$ .

**Proposition 5** highlights the nuance in the degree of concentration across market structures. As discussed in the proof sketch of **Theorem 2**,  $c_g$  is a sufficient statistic for the order in which different genres are activated, in both competitive equilibria and the platform's allocation.<sup>9</sup> However, **Proposition 5** shows that once a genre is active, how much attention it receives depends finely on the distribution of demand within that genre, and the market structure. Competitive equilibria rewards genres where many producers consume a small amount of the good, and those in which consumers are similar to one another. The platform's behaviour is more nuanced: it behaves like the competitive equilibrium when its resources  $m$  are small, but directs attention towards genres with many "super-fans", in order to benefit from their consumption of a large amount of content.

## 6.2 The role distribution of realized quality $F$ .

By **Theorem 1** and **Proposition 3**, uncertainty over quality is necessary and sufficient for allocative inefficiency to exist. It is hence important to understand how changes in this underlying uncertainty affect the allocation of producers in different market structures. An important feature of the distribution of quality realization that will be important is the weight of the tails of the distribution. In this section, I show that when the right tail of  $F$  is thin, the platform creates a more diverse library, relative to when the tails of  $F$  are thick.

At a high level, when the tails of  $F$  are thin,  $\bar{F}^{-1}\left(\frac{t}{\psi_g}\right)$  converges quickly, as a function of  $\psi_g$ . When  $F$  has thicker tails,  $\bar{F}^{-1}\left(\frac{t}{\psi_g}\right)$  converges more slowly. When the tails of  $F$  are small, in its tails, an extremely large increase in  $\psi_g$  is necessary to move the upper quantile of the measure by a small amount. On the other hand, if the tails of  $F$  are thick, even in the tails of the distribution there is significant mass, so that the upper quantiles are far apart.

The translation into how the platform incorporates tail weight is as follows. Consider increasing  $m$  from zero. The platform first focuses on the genre  $g$  that has the

---

<sup>9</sup>By order, I mean as a function of increasing the mass of producers  $m$ .



largest overall demand  $c_g$ . The platform continues to invest all producers in this genre until the benefit from increasing  $\psi_g$  decreases to the point that other genres become sufficiently attractive. When the tails of  $F$  are thin, this decrease happens quickly by the above observation: consumer value thresholds increase slowly, so that their utility increase from increasing  $\psi_g$  is small. On the other hand, if the tails of  $F$  are thicker, this benefit decreases more slowly: there is sufficient mass in the tails for the platform to benefit from continuing to increase  $\psi_g$ . This suggests that as the tails of  $F$  increase in mass, the platform concentrates its effort on fewer genres.

In the competitive equilibrium, the effect depends dramatically on how sensitive pricing is to realized quality. If pricing is flat, as long as  $\bar{F}(p)$  doesn't change, changing the weight of the tails has no effect on producer incentives. However, if price increases with quality, a similar force encourages producers in the same direction as the platform: the benefit of high-quantile realizations increases, increasing concentration.

These results are summarized in the following proposition. In order to capture the effects of increasing the tail weight of the distribution, I consider a transformation to quality that “spreads-out” its realizations: a transformation  $\tau(v)$  such that  $\tau$  is convex. In what follows, let  $A_\tau^S$  denote the active set in market structure  $S$  following the transformation  $\tau$  (and similarly  $\psi_\tau^S$ ).

**Proposition 6.** *Fix a transformation of quality  $v \mapsto \tau(v)$  with  $\tau$  increasing and  $\tau(0) = 0$ , such that  $F_\tau$  also has increasing hazard rates. Let  $P$  be an admissible pricing function with  $\underline{v}_P = 0$ . Then*

- (i) *If  $\tau$  is convex, product range decreases:  $A^I \supseteq A_\tau^I$  and  $A^P \supseteq A_\tau^P$ .*
- (ii) *If  $\tau$  is concave, product range increases:  $A^I \subseteq A_\tau^I$  and  $A^P \subseteq A_\tau^P$ .*
- (iii) *This change has no effect on the  $p = 0$  competitive equilibrium:  $\psi^0 = \psi_\tau^0$ .*

The assumption that  $\tau(0) = 0$  is to ensure that  $\bar{F}(0) = \bar{F}_\tau(0)$ .

**Proposition 6** highlights an interesting tension between flat pricing and price-dependent pricing. As observed in **Proposition 4**, allowing firms to differentiate based on price drives an additional wedge between the competitive equilibrium and the platform-optimal allocation. This result shows that this isn't due to producers closely

mimicking the platform when pricing is flat. Instead, flat-pricing simply ensures that producers are largely agnostic about underlying uncertainty. On the other hand, when producers do internalize the distribution of quality, their behaviour changes in the same way as the platform’s in response to changes in underlying uncertainty.

## 7 Recommendation System

In many settings, platforms act as intermediaries without directly coordinating production. These intermediaries act as *recommendation systems*, matching consumers with content for them to consume. In this section I show how recommendation systems’ ability to artificially suppress content can influence producer incentives in order to increase overall welfare.

In my setting, a recommendation system is simply a black-box which connects consumers to products. In the baseline competitive equilibrium, there are no informational frictions: consumers are always able to find and consume the content that gives them the highest utility. Hence, unlike in other models of recommendation systems where the primary purpose of recommendation systems is to expose consumers to products that they wouldn’t otherwise see, in my model the recommendation serves the opposite role.

Recall the motivating example of [Section 2](#). By committing to selectively *not* recommend (i.e., suppress) some content, the platform influences producer incentives. This allows the recommendation system to indirectly shift the allocation of producers across genres. Due to the inefficiency of the competitive equilibrium allocation of producers, it is possible that the suppression of content—which has a first-order effect of destroying surplus—could increase overall welfare by resulting in a more efficient allocation of producers across genres. Notice, however, that there is scope for a recommendation system to influence production in this way only if there is ex-ante uncertainty over quality. If producers knew their quality ex-ante, then any suppression of content would result in a decrease in overall welfare, since they are already sorting efficiently.

In my setting, a recommendation system is simply a collection of “sub-genres”  $\mathcal{S}$ . Each sub-genre  $s$  consists of (i) a genre  $g_s$ , to which content produced in the sub-genre belongs, and (ii) a recommendation function  $r_s : \mathbb{R} \times \mathbb{R}_+ \rightarrow [0, 1]$ . The recommendation function  $r_s(v, t)$  is the probability of a consumer of type  $(g_s, t)$  is

shown a specific piece of content with quality  $v$  from the sub-genre. In order to map directly into settings where these recommendation systems persist, I assume that producers generate profits proportional to the mass of consumers that consume their content.<sup>10</sup>

The timing is as follows:

1. The platform commits to a recommendation system  $\mathcal{S}$ .
2. Producers make production choices, inducing  $\tilde{\psi} \in m\Delta\mathcal{S}$ .
3. Producer quality is realized, and the platform bundles goods and prices bundles according to the recommendations from  $r_s$ .
4. Consumers consume those products in their bundle that given them the highest utility, and produce profits are realized.

I again allow the platform to bundle goods and price according to the utility of these bundles. In this case, the recommendation system is able to extract full surplus from consumers, subject to incentive compatibility constraints on producers, and the recommendation functions  $r_s$ . In this way, the problem of the recommendation system becomes that of a constrained social planner.

**Definition 8.** A recommendation system is a set of sub-genres  $\mathcal{S}$  and an allocation  $\tilde{\psi} \in m\Delta\mathcal{S}$  such that

$$\sum_{g \in G} \int_0^\infty \left( \int_{\max\{v_g(t), 0\}}^\infty v d\mu_{g,t}(v) \right) dT_g(t)$$

such that

$$\begin{aligned} d\mu_{g,t}(v) &= \sum_{s=(g, r_s)} \tilde{\psi}_s r_s(v, t) f(v) \\ t &= \mu_{g,t}([v_g(t), \infty)) \\ \tilde{\psi}_s > 0 &\iff s \in \operatorname{argmax}_{s \in \mathcal{S}} \int_0^\infty \left( \int_{\max\{0, v_g(t)\}}^\infty r_s(v, t) f(v) dv \right) dT_g(t) \end{aligned}$$

---

<sup>10</sup>Spotify, for instance, pays producers based off of their “stream-share”, which due to the large number of total streams, effectively grows linearly in an artist’s own streams.

The platform aims to maximize the total utility of consumers in its platform, subject to the constraints that (i) consumers are only able to consume content that they are recommended, and (ii) producers produce the content that yields the highest expected profits.

While the space of recommendation systems is large (as  $\mathcal{S}$  is arbitrary and chosen by the platform), optimal recommendation systems are straightforward.

**Lemma 3.** *Any optimal recommendation system is outcome-equivalent to a recommendation system containing one sub-genre for each genre, where  $r_s(v, t) = \mathbb{1}\{v \geq p_g\}$  if  $g_s = g$ , for some  $p_g \in \mathbb{R}_+$ . That is, the problem of finding an optimal recommendation system is equivalent to solving the following problem:*

$$\begin{aligned} \max_{p_g, \psi} \sum_{g \in G} \psi_g \cdot \int_0^\infty W\left(\min\left\{\bar{F}(p_g), \frac{t}{\psi_g}\right\}\right) dT_g(t) \\ \text{s.t. } \psi_g > 0 \iff g \in \operatorname{argmax}_{g \in G} \int_0^\infty \min\left\{\bar{F}(p_g), \frac{t}{\psi_g}\right\} dT_g(t) \end{aligned}$$

Lemma 3 shows that every optimal recommendation system looks like a recommendation system where the platform defines a *genre-specific* minimum threshold of content, and recommends any content above this minimum threshold to all consumers. Hence, the optimal suppression of content takes the form of content curation: it may be difficult to be allowed onto the platform, but once a product is accepted it becomes available to all consumers. Equipped with this straightforward (albeit nuanced) maximization problem, I now state the major result of this section.

**Theorem 3.** *Fix an optimal mechanism  $(p^R, \psi^R)$  and the corresponding active set  $A^R = \{g | \psi_g^R > 0\}$ .*

(i) *It is possible for the optimal mechanism to strictly improve on the  $p = 0$  competitive equilibrium;*

(ii)  *$A^{0,0} \subseteq A^R$ ; and*

(iii)  *$A^R \subseteq A^I$ .*

*Both inclusions in (ii) and (iii) may be strict.*

The intuition for this result is the same as the intuition from the motivating example. By suppressing content in a high-demand genre, producers’ incentives may disproportionately shift them towards producing in a smaller genre. Due to the inefficiency of the original sorting, this benefit may be large enough to offset the loss from suppressed content.

## 8 Conclusion

Entertainment products are ubiquitous, and important for consumer welfare. Understanding how prevailing market structures interact with producer incentives is integral to understanding and predicting how the distribution of products will evolve as market structures continue to change. I construct a straightforward model of producer incentives and show that ex-ante uncertainty over own quality drives an allocative market failure: producers sort across product types inefficiently. I show that this allocative inefficiency manifests in the form of increased concentration: there is too little diversity in products. Integrating producers into a single platform alleviates this market inefficiency at the cost of consumer surplus. I also show that a recommendation system is able to overcome some of this inefficiency by selectively suppressing some content. Specifically, the recommendation system suppresses content in high-demand genres in order to encourage producers to explore the genres inefficiently ignored in competitive equilibrium.

## References

- Aguiar, L. and J. Waldfogel (2016): “Even the losers get lucky sometimes: New products and the evolution of music quality since Napster”. *Information Economics and Policy* 34, pp. 1–15 (cit. on p. 3).
- (2018): “Quality predictability and the welfare benefits from new products: Evidence from the digitization of recorded music”. *Journal of Political Economy* 126(2), pp. 492–524 (cit. on pp. 1, 4).
- (2021): “Platforms, power, and promotion: Evidence from Spotify playlists”. *The Journal of Industrial Economics* 69(3), pp. 653–691 (cit. on p. 4).
- Aguiar, L., J. Waldfogel, and S. Waldfogel (2021): “Playlisting favorites: Measuring platform bias in the music industry”. *International Journal of Industrial Organization* 78, p. 102765 (cit. on p. 4).

- Calvano, E., G. Calzolari, V. Denicolo, and S. Pastorello (2025): “Artificial intelligence, algorithmic recommendations and competition”. *Algorithmic Recommendations and Competition (February 19, 2025)* (cit. on p. 4).
- Chen, Y., F. Li, and M. Preuss (2025): “Algorithmic attention and content creation on social media platforms”. *Available at SSRN* (cit. on p. 4).
- Deng, X., N. Li, W. Li, and Q. Qi (2025): “Competition among parallel contests”. *Information and Computation*, p. 105339 (cit. on p. 4).
- Drugov, M. and D. Ryvkin (2020): “Tournament rewards and heavy tails”. *Journal of Economic Theory* 190, p. 105116 (cit. on p. 4).
- Gambato, J. and L. Sandrini (2025a): “Dance to my tune! Discovery mode and built-in recommendation bias”. *Discovery Mode and Built-in Recommendation Bias (July 15, 2025)*. ZEW-Centre for European Economic Research Discussion Paper (25-037) (cit. on p. 4).
- (2025b): “Not as good as it used to be: Do streaming platforms penalize quality?” *Working Paper* (cit. on p. 4).
- Grossmann, M. (2021): “Entry regulations and optimal prize allocation in parallel contests”. *Review of Economic Design* 25(4), pp. 289–316 (cit. on p. 4).
- Hennig-Thurau, T., M. B. Houston, et al. (2019): “Entertainment science”. *Data Analytics and Practical Theory for Movies, Games, Books, and Music*. Cham: Springer Nature (cit. on pp. 1, 6, 10, 20).
- Lee, K. H. and L. Musolff (2025): “Two-sided markets shaped by platform-guided search”. *Working Paper* (cit. on p. 4).
- Piolatto, A. and F. Schuett (2012): “Music piracy: a case of “the rich get richer and the poor get poorer””. *Information Economics and Policy* 24(1), pp. 30–39 (cit. on p. 3).
- Ryvkin, D. and M. Drugov (2020): “The shape of luck and competition in winner-take-all tournaments”. *Theoretical Economics* 15(4), pp. 1587–1626 (cit. on p. 4).
- Waldfogel, J. (2013): “Cinematic Explosion: Movies, Gatekeepers, and Product Discovery in the Digital Era”. *Working Paper* (cit. on p. 3).